Akash Reddy Vangala

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PROFESSIONAL SUMMARY

Recent Computer Science graduate with hands-on experience developing C++ applications using object-oriented programming, STL containers, smart pointers, and modular software design. Strong foundation in data structures, algorithms, debugging, and problem-solving, with practical experience building interactive applications and graphical systems using SFML. Interested in high-performance computing, efficient software architecture, and low-latency trading technologies. Eager to collaborate with experienced engineers and develop reliable, performance-focused C++ systems.

EDUCATION

University of Massachusetts Lowell May 2026
Bachelor of Science in Computer Science Lowell, MA

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, SQL, PHP

C++ Development: OOP, STL, Smart Pointers, Memory Management, JSON, SFML

Core Computer Science: Data Structures, Algorithms, Debugging, Problem Solving, Software Design

Developer Tools: Git, GitHub, Linux, G++, GDB, VS Code, Docker, CI/CD

PROJECTS

AniPlayer – Modular 2D Animation Engine 2026
C++, SFML, JSON, STL, Object-Oriented Programming

- Designed and developed a modular 2D animation engine in C++ using SFML, rendering configurable graphical objects from JSON configuration files.
- Implemented keyframe-based tweening, scaling, rotation, audio playback, and interactive controls for dynamic animation sequences.
- Applied object-oriented design, STL containers, and smart pointers to organize reusable components and manage graphical resources.

Sokoban – Interactive Puzzle Game 2026
C++, SFML, STL, Object-Oriented Programming

- Developed an interactive puzzle game in C++ using SFML, separating rendering, user input, and game logic into modular components.
- Implemented collision detection, movement validation, box-pushing mechanics, undo/redo operations, and game-state management.
- Used STL containers and object-oriented programming to organize game objects, track player actions, and manage puzzle states.

SpriteBuilder – Interactive Graphics Application 2026
C++, SFML, Object-Oriented Design

- Developed a C++ graphics application using SFML to create and manipulate interactive 2D graphical objects.
- Implemented event-driven input handling and graphical transformations for interactive object movement and rendering.
- Organized application components using object-oriented principles to separate graphics behavior and user interaction.

Mock University – Full-Stack Management Application Feb 2026 – Apr 2026
Java, PHP, SQL, Android, Git

- Collaborated in a three-member team to build a university management application supporting authentication, registration, and academic records.
- Designed relational database schemas and SQL queries; developed backend API endpoints for retrieving and updating application data.
- Implemented Android features in Java and collaborated using Git to maintain application functionality and source code.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** – Amazon Web Services, Jan 2026
- **KodeKloud Engineer – Kubernetes** – May 2026
- **KodeKloud Engineer – 100 Days of Cloud (AWS)** – Feb 2026